

Safe inference and the stability-controlled quasi-experiment (SCQE)

PS200C, Spring 2026

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Roadmap

- 1 Motivation: “safe inference” as a posture
- 2 The SCQE idea and identification
- 3 Applications: dexamethasone, HCQ, remdesivir
- 4 Relationship to other strategies (DID, SOO+ sensitivity)
- 5 Wrap-up

Motivation

Many high-stakes treatments get rolled out, taken, or assigned without randomization.

For example: in the early pandemic, many people were given unproven therapies outside of RCTs. **What could we have learned from those experiences?**

Reaction 1: “If it isn’t an RCT, it is rubbish.”

- Healthy starting point...point estimate cannot be trusted...but going beyond the heuristic you might find reliable information.

Reaction 2: “Make a heroic assumption, call the estimate suggestive.”

- Covariate adjustment + no-unobserved-confounders is the classic case, but the same goes for IV, DID, synth, etc.
- Even calling the results “suggestive” is dangerous... helpful treatments may look harmful, vice versa.

An alternative: safe inference

Safe inference: partial identification + sensitivity analysis.

- Pick an identification strategy whose **assumptions you are equipped to argue about**.
- Argue what **bounds** you can defend on those assumptions, see what conclusion they buy you.
- Report ranges of estimates indexed by the assumption, not a single point estimate that hides the assumption.

We did something of this kind with SOO + sensitivity.

Today: a different identification strategy, designed to be argued about transparently.

The stability-controlled quasi-experiment (SCQE)

Example: COVID-19 patients admitted to a hospital in Feb–Mar 2020.

- Of patients admitted in February, 0% got dexamethasone
- Of patients admitted in March, 30% (very non-randomly) received it

Ask yourself, absent this treatment how much could the mortality rate change from one month to the next? I.e. the “baseline trend”.

$$\delta \equiv \mathbb{E}[Y(0)|T = 1] - \mathbb{E}[Y(0)|T = 0]$$

This baseline trend δ is the only assumption required to get the ATT.

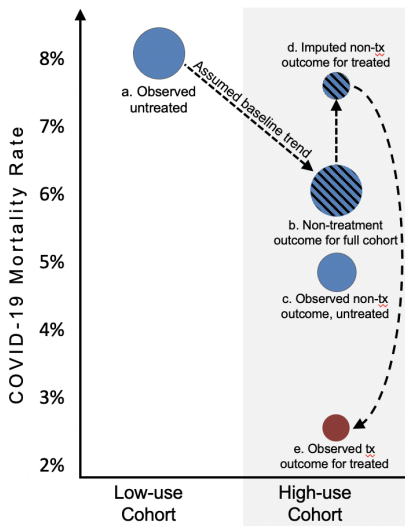
SCQE: Arguments about δ

We'll come back to this in practice but to set the stage: You don't know δ and it doesn't live in the data. But, you can reason about it through:

- Hard bounds (mortality rate between 0 and 1)
- Reasonable (disputable) bounds: mortality almost surely didn't double, or halve, from one month to the next; if it had we would know...
- Informed bounds: based on observed changes in patient characteristics, other treatments, and expert judgment, you might guess that mortality (absent dexamethasone) likely dropped by...

Don't worry we won't need point values for δ ...there will be good uses for (i) defensible ranges, (ii) unbelievable ranges, or even just (iii) ranges where you cannot be sure either way...all will contribute to characterizing what we can(not) conclude.

How does δ +data tell you the ATT?



- We observe (a), $\mathbb{E}[Y(0)|T = 0]$. For a postulated δ , you get out (postulated) $\mathbb{E}[Y(0)|T = 1]$ (b).
- Meanwhile in later cohort, you observe $\mathbb{E}[Y(0)|T = 1, D = 0]$ (c).
- Then you can back out $\mathbb{E}[Y(0)|T = 1, D = 1]$ (d).
- We observe $\mathbb{E}[Y(1)|T = 1, D = 1]$ (e), and that minus (d) is the ATT.

Mathified,

Define the change in non-treatment average outcomes,

$$\delta \equiv \mathbb{E}[Y(0)|T = 1] - \mathbb{E}[Y(0)|T = 0]. \quad (1)$$

By LIE,

$$\mathbb{E}[Y(0)|T = 1] = \mathbb{E}[Y(0)|T = 1, D = 1]\pi_1 + \mathbb{E}[Y(0)|T = 1, D = 0](1 - \pi_1)$$

So we have

$$\mathbb{E}[Y(0)|T = 0] + \delta = \mathbb{E}[Y(0)|T = 1, D = 1]\pi_1 + \mathbb{E}[Y(0)|T = 1, D = 0](1 - \pi_1) \quad (2)$$

$$\mathbb{E}[Y(0)|D = 1, T = 1] = \frac{\mathbb{E}[Y|T = 0] - \mathbb{E}[Y|D = 0, T = 1](1 - \pi_1) + \delta}{\pi_1} \quad (3)$$

So if you want the ATT at $T = 1$,

$$\begin{aligned} ATT &\equiv \mathbb{E}[Y(1)|D = 1, T = 1] - \mathbb{E}[Y(0)|D = 1, T = 1] \\ &= \mathbb{E}[Y|D = 1, T = 1] - \left(\frac{\mathbb{E}[Y|T = 0] - \mathbb{E}[Y|D = 0, T = 1](1 - \pi_1) + \delta}{\pi_1} \right). \quad (4) \end{aligned}$$

Some details

- Works the same way when there are treatment-takers in both periods — what you need is a **big change in treatment probability**. (You then get a LATE-like quantity rather than the ATT.)
- The two “cohorts” could be different facilities, eras, or sites — whatever defines a sharp shift in treatment probability, as long as you can reason about non-treatment outcome differences between them.
- In fact you could do this with just one study cohort, and any guess of $\mathbb{E}[Y(0) \mid \text{your study sample}]$ (more soon)

Connection to IV: time as the instrument

Cohort/time is like an encouragement to treatment:

- If no one is treated in the first cohort, you have one-way non-compliance (no always-takers), so the complier-ATE is just the ATT.
- **But:** the exclusion restriction is not expected to hold here. Time affects mortality through many other channels... that is what δ is about!
- The math effectively subtracts the exclusion restriction violation off,

Concretely: define a pseudo-outcome $\tilde{Y}_i \equiv Y_i - \delta T_i$. Then under SCQE assumptions $\tilde{Y}_i(0) \perp\!\!\!\perp T_i$, and the Wald estimator gives:

$$ATT = \frac{\mathbb{E}[Y|T=1] - \mathbb{E}[Y|T=0] - \delta}{\pi_1}.$$

...which is just a (simpler looking) rearrangement of the SCQE estimator in (4).

Worked example: dexamethasone for COVID-19

Cohorts:¹

- “Low-use” (first) cohort: days 44–101, **5.7%** took dexamethasone
- “High-use” (second) cohort: days 102–200, **46%** took dexamethasone

Outcome: 14-day mortality

- **10.9%** in the low-use (first) cohort
- **5.4%** in the high-use (second) cohort

On its surface: big uptick in treatment, big drop in mortality...promising but could it just be the baseline trend?

¹Wulf, Hazlett, Hill, Chiang, Goodman-Meza, Pasaniuc, Arah, Erlandson, Montague. “Safe inference outside of randomized trials: Application of the stability-controlled quasi-experiment to the effects of three COVID-19 therapies.” Observational Studies 11(3), 2025, 301–330.

Dexamethasone: δ considerations

Before looking at the SCQE results, let's think about where δ might be.

Baseline characteristics. High-use cohort had fewer ICU admissions, ventilator use, skilled-care admits within 24h — all consistent with a drop in baseline mortality.

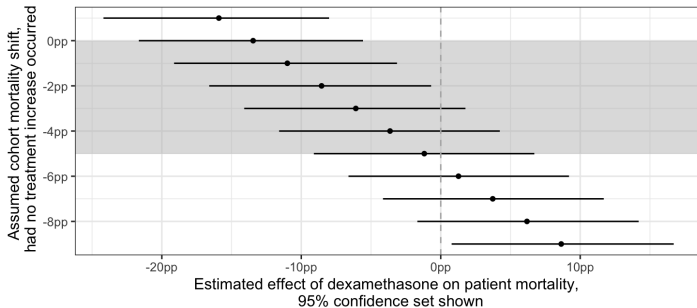
Other therapies. Higher remdesivir use in the high-use cohort (12% vs. 27%). Could reduce risk; unknown at the time.

Modeled mortality risk on observables. Predicted baseline mortality fell by 1.3–2.1pp.

Expert judgement. Physicians on the team expected baseline mortality to drop with generally improving practice over this period.

“All lines of evidence suggest a possible drop in baseline mortality. Taking these considerations collectively, and maintaining a conservative attitude, we postulated that a plausible range for the baseline trend ranged from no change to a reduction of up to 5pp (–46%).”

Dexamethasone: results



- Beneficial ($p < 0.05$) if baseline mortality increased, flat, or fell by up to 2.3pp (−21%)
- Non-harmful point estimate until a drop of 5pp (−46%)
- Harmful ($p < 0.05$) only if baseline mortality fell by 6.8pp (−93%), leaving mortality of just 0.4%

Dexamethasone: what we concluded

“We consider [the drop in baseline required to support a harmful effect] to be extremely unlikely, particularly without a known cause that escaped the attention of physicians on the team and does not appear in the observable differences in cohorts.”

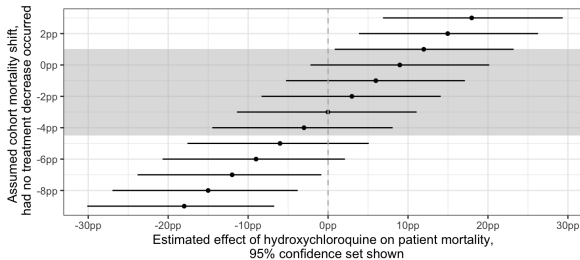
“Thus, we have considerable confidence that dexamethasone was at least directionally, and possibly statistically significantly beneficial, and that it was not significantly harmful.”

Consistent with the eventual results of the RECOVERY-trial.

Hydroxychloroquine (HCQ): results

Cohort shift: 36% HCQ use in high-use early cohort → 2.9% in low-use cohort. Mortality 11.6% → 8.6%.

Observables/ expertise suggested: “a wide range of baseline trends are possible, from +1pp to −4.5pp.”



Wide plausible range of δ in which you would conclude HCQ was **harmful**.

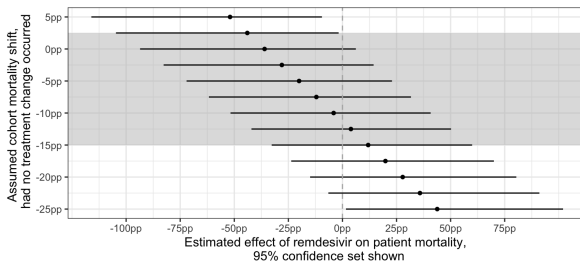
Sig. harmful if baseline mortality worsened by even 0.7pp.

Sig. beneficial only if baseline improved by $\geq 6.7\text{pp}$ (−46%): not credible.

Remdesivir: results

Cohort: 0% remdesivir use in no-use cohort → 31.3% in high-use cohort.
Mortality 24.5% → 13.3%.

Observables and clinical judgment: plausible range $\delta \in [+2.5\text{pp}, -15\text{pp}]$.



While benefit is plausible, cannot rule out harm (on point estimates).

Can rule out statistically significant harm – would imply 0% counterfactual mortality.

Conclusions

- **Dexamethasone:** likely beneficial or null. Harm would require an implausibly large drop in baseline mortality.
- **HCQ:** Could be (sig.) harmful with small positive δ . Sig. benefit would require larger drop than deemed plausible.
- **Remdesivir:** sig. benefit is plausible but so is harmful (point) estimate. Not sig. harmful.

If you had to decide what to do, you'd have some guidance – it would take insane assumptions to believe HCQ will help, whereas dexamethasone possibly could; neither dexamethasone nor remdesivir can reasonably be significantly harmful, but HCQ can be.

Not precise, but a long way from either “we can't say anything without an RCT” or “the observational study is our best guess.”

Relationship to DID

Case 1: You can't do DID at all.

- DID requires labeling units pre-treatment as “would-be-treated” vs. control.
- Impossible in cases like this – you cannot label patients in the first cohort as “would-be treated”.

Case 2: You can define those groups.

- Parallel trends = the two groups share a trend in $\mathbb{E}[Y(0)]$.
- Suppose you just look at change in mean outcomes in the control group, and you assume that is exactly δ .
- That is DID! So DID is a special case of SCQE, with a particular assumption about δ .
- SCQE shows DID as one estimate in a range, indexed by δ .

Relationship to covariate adjustment + sensitivity

What does SCQE reveal that a sensitivity-analyzed covariate-adjustment analysis wouldn't?

Direct point-estimate comparison is not meaningful here.

Instead, compare what conclusions you could defend through each:

- Across these COVID applications, unobserved confounding explaining $\approx 18\%$ of residual variance in treatment and outcome would suffice to flip the significance of any conclusion under covariate adjustment.
- We can't really rule that out...so we can't rule out any conclusions via SOO+sensitivity.

Different assumptions offer different leverage... use what works.

Software

R package `scqe`: <https://github.com/chadhazlett/scqe>

- Two-cohort and one-cohort modes (full data or summary stats)
- ATT estimates and CIs over a range of δ
- Anderson–Rubin intervals and block-bootstrap CIs

Browser calculator: <https://chadhazlett.com/scqe-app/>

- Type in cohort-wise summary stats, get the SCQE plot back
- Both two-cohort and one-cohort tabs
- Includes the dexamethasone worked example as a built-in case

You can plug in just cohort-by-treated numbers; doesn't require individual data..

Where to use SCQE

Medicine:

- While we wait for RCTs — present observational results in ways that prevent over-confidence
- Therapies unlikely to be tested in trials
- Post-approval real-world effectiveness, with different populations / protocols than trials
- Prospective observational designs aimed at reducing δ

Outside medicine: anywhere there's a sharp change in treatment likelihood and you can't lean on no-confounding or parallel trends:

- rolling out a new feature people adopt by choice
- effects of new classes / training programs / curricula opted into
- media or messaging that people consume without randomization

Takeaways

Safe inference = extract what you can with assumptions you can defend, instead of selling false certainty around a point estimate.

- SCQE: one assumption, δ (the baseline trend), out in the open.
- Different ID strategies expose different assumptions to argument — **use what you've got**.
- SCQE can do work where DID and IV can't (no labeled control group, no clean exclusion).
- The one-cohort form gives you a partial-ID treatment of external control arms.
- Comfort with partial identification takes practice. Reporting ranges instead of points is the point.

Appendix: inference

Consider the regression using the pseudo-outcome $\tilde{Y}_i = Y_i - \delta T_i$:

$$\tilde{Y}_i = \beta_0 + \beta_{\text{SCQE}} D_i + \mu_i,$$

where β_{SCQE} is the SCQE estimate.

Under spherical errors,

$$\widehat{SE}(\hat{\beta}_{\text{SCQE}}) = \frac{\hat{\sigma}_\mu}{\sqrt{N} \hat{\rho}_{D,T} \hat{\sigma}_D},$$

where $\hat{\rho}_{D,T}$ is the sample correlation of treatment and time indicators.

More generally use the IV sandwich:

$$\widehat{\text{Var}}(\hat{\beta}_{\text{SCQE}}) = (\mathbf{Z}^\top \mathbf{D})^{-1} \mathbf{Z}^\top \hat{\mathbf{\Pi}} \mathbf{Z} (\mathbf{Z}^\top \mathbf{D})^{-1}$$

with $\mathbf{Z} = [1 \ T]$, $\mathbf{D} = [1 \ D]$, and $\hat{\mathbf{\Pi}}$ a consistent estimator for $\text{Var}(\mu)$ (e.g. HC2).

Alternatives in the package: Anderson–Rubin intervals (no weak-instrument concerns), block-bootstrap over clinics for cluster-robust inference.